IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 14 July 2026, accepted 18 August 2026, date of publication 24 August 2026, date of current version 9 September 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3727143

RESEARCH ARTICLE

Explainable Anomaly Detection in Accounting Journal Entries Using Autoencoders and Counterfactual Explanations

YOHEI KAWAZURA[1,2] **AND DAISUKE SAITOU**[2,3]

[1]School of Data Science and Management, Utsunomiya University, Utsunomiya, Tochigi 321-8505, Japan
[2]Center for Data Science, Utsunomiya University, Utsunomiya, Tochigi 321-8505, Japan
[3]TKC Corporation, Utsunomiya, Tochigi 320-8644, Japan

Corresponding author: Yohei Kawazura (kawazura@a.utsunomiya-u.ac.jp)

This work was supported by TKC Corporation.

ABSTRACT In financial auditing, an autoencoder (AE) can flag a journal entry as anomalous, but the auditor still needs to know which attributes triggered the flag and how to correct the entry. We address both questions with a pipeline that uses attribute-level SHAP (SHapley Additive exPlanations) attribution (RESHAPE) for root-cause identification, and RESHAPE-guided beam search with gradient-based numeric optimization for counterfactual explanation (CF) generation. We use two thresholds: global binary cross-entropy (BCE) loss for detection, and per-attribute thresholds for CF validity. The detection threshold is selected label-free, from a target false-positive rate measured on a held-out clean validation set, so that no anomaly labels enter threshold selection. On 532,789 normal SAP-format accounting journal entries, split into training, validation, and test partitions with 120 injected anomalies held out for evaluation, the detector flagged all 120 anomalies (100% recall) while raising only 31 false positives on the 79,919-entry clean test set (false-positive rate 3.9×10^{-4}). On those 120 anomalies, the CF pipeline exactly matched the ground-truth modified attributes in 30 cases (25.0%) and hit at least one ground-truth attribute in 99 cases (82.5%), averaging 1.18 modifications per CF, and outperformed the Wachter, DiCE, and FACE counterfactual baselines, achieving the highest exact-match rate at full validity with the sparsest corrections.

INDEX TERMS Anomaly detection, autoencoder, counterfactual explanation, accounting journal entries, explainable AI.

I. INTRODUCTION

Detecting fraudulent or erroneous journal entries is a persistent problem in financial auditing. Autoencoder (AE) based anomaly detection is a natural fit because it can be trained on a curated set of normal transactions, without ever seeing labeled anomalies, and then flag entries that the model fails to reconstruct [1], [2], [3].

A binary anomaly label is not enough for the auditor. The auditor also needs *why* the transaction is anomalous, for the audit report, and *what to change* to make it normal, for the correction itself. Existing AE-based detection usually relies on a single scalar reconstruction error (binary cross-entropy, BCE), which does not break down by attribute. Most counterfactual explanation (CF) research targets classifiers [4] and does not deal with AE-specific issues like per-attribute reconstruction error thresholds or search over mixed discrete-continuous spaces. SHAP (SHapley Additive exPlanations) [5] gives attribute-level attribution for AE reconstruction errors, but attribution alone does not tell the auditor what to write in the corrected entry.

Building on the AE-based anomaly detection framework of Schreyer et al. [1], this study makes the following contributions:

1) **RESHAPE-guided beam search**: Attribute mutations are ordered by RESHAPE (Reconstruction Error

The associate editor coordinating the review of this manuscript and approving it for publication was Ikramullah Lali.

© 2026 The Authors. This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

SHapley Additive exPlanations) [6] attribution priorities, with per-attribute mutability constraints.
2) **Gradient-based numeric optimization**: Post-hoc refinement of numeric attributes via gradient descent to find sample-specific AE reconstruction error minima.
3) **Two-stage thresholding**: Global BCE loss thresholding for detection and per-attribute thresholding for CF validity.
4) **Systematic evaluation**: Assessment across four anomaly subtypes (see Section III-B for details) covering detection, CF generation, and ground-truth alignment, with a label-free threshold-selection protocol, an ablation study, a comparison against counterfactual baselines, a hyperparameter-sensitivity analysis, and a false-positive analysis.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the problem setting and data. Section IV details the proposed method. Section V presents experimental results. Section VI discusses the findings and limitations, and Section VII concludes.

II. RELATED WORK

A. ANOMALY DETECTION IN ACCOUNTING DATA

Traditional anomaly detection in accounting relied on rule-based methods (amount thresholds, posting pattern matching), which miss patterns the rules do not anticipate. Schreyer et al. [1] applied deep autoencoders to accounting journal entries and showed that reconstruction error works well as an anomaly score. This line of work was extended to adversarial AEs [3], multi-view contrastive learning [7], and federated learning [8]. Schultz and Tropmann-Frick [2] compared AE-based detection with external auditor judgments and found reasonable agreement. Hemati et al. [9] studied continual learning for continuous auditing scenarios. Sattarov et al. [10] proposed attribute-level anomaly scoring using denoising AEs (attribute-level attribution methods are compared in Section II-C).

Except for Sattarov et al. [10], prior work relies on aggregate reconstruction error without providing attribute-level explanations. Torabi et al. [11] demonstrated the effectiveness of per-attribute thresholds based on vector reconstruction error, but did not apply this to accounting data or integrate it with counterfactual explanations. Our work combines global BCE thresholding for detection with per-attribute thresholding for CF quality assessment, supporting both root-cause identification and the correction step.

B. COUNTERFACTUAL EXPLANATIONS

Counterfactual explanations answer the question "what feature changes would alter a model's decision?" [4], [12], [13].

Wachter et al. [14] proposed gradient-descent-based CF generation by minimizing a loss that trades off the desired output against proximity to the original input; discrete tabular attributes are handled only afterwards, by rounding the continuous solution. DiCE [15] adds diversity constraints to produce multiple diverse CFs simultaneously, sampling and filtering candidates against the classifier rather than exploiting per-attribute structure. FACE [16] searches for CFs along graph shortest paths on the data manifold, returning an existing high-density point and so guaranteeing feasibility at the cost of minimality. CERTIFAI [17] uses genetic algorithms for model-agnostic CF generation. Growing Spheres [18] progressively expands random perturbation spheres to find CFs near the decision boundary. DACE [19] formulates CF generation as an optimization problem with integer constraints to ensure realistic outputs. Karimi et al. [20] framed CFs as algorithmic recourse, incorporating causal relationships between features.

For anomaly detection, Haldar et al. [21] proposed reliable CF generation for AE-based anomalies, and Sulem et al. [22] explored diverse CF generation for time-series anomaly detection. However, to our knowledge, these methods do not combine search over mixed discrete-continuous spaces with normality assessment based on per-attribute reconstruction error.

Our method differs from these approaches in three ways: (1) a two-stage design using global BCE loss for detection and *per-attribute reconstruction error* for CF validity; (2) RESHAPE (SHAP attribution) for *search priority guidance*; (3) combining beam search with *gradient-based numeric optimization* for efficient exploration of mixed discrete-continuous spaces. Section V-E reports a direct empirical comparison against Wachter, DiCE, and FACE.

C. SHAP AND EXPLAINABLE AI

SHAP (SHapley Additive exPlanations) [5] is a feature attribution framework grounded in game-theoretic Shapley values; it includes the model-agnostic KernelExplainer. Antwarg et al. [23] applied SHAP to AE anomaly detection and found that SHAP attribution identifies true anomaly causes better than raw reconstruction error. Mueller et al. [6] specialized this for accounting audits as RESHAPE (Reconstruction Error SHapley Additive exPlanations), which aggregates feature-level SHAP values to the attribute level. We use RESHAPE attribution values as the attribute search priorities in CF generation.[1]

Li et al. [24] surveyed explainable anomaly detection, categorizing feature attribution, counterfactual, and rule-based approaches. In auditing specifically, Zhang et al. [25] applied LIME (Local Interpretable Model-agnostic Explanations)

[1] Sattarov et al. [10] also decompose AE reconstruction error by attribute for anomaly attribution, but compute confidence scores directly from denoising AE output probabilities. RESHAPE instead uses SHAP's KernelExplainer to compute Shapley values, which guarantees axiomatic properties (local accuracy, missingness, consistency) [5]. More importantly, Sattarov et al. [10] treat anomaly attribution as the final output, whereas we feed RESHAPE attribution into CF search priorities, connecting "which attribute is anomalous" to "what change would normalize it." In principle, the confidence scores of Sattarov et al. [10] could also serve as CF search priorities.

and SHAP to audit risk assessment, and Černevičienė and Kabašinskas [26] reviewed explainable AI (XAI) research in finance.

D. POSITIONING OF THIS WORK

Table 1 positions the proposed method against representative AE-based auditing methods, attribute-level/RESHAPE attribution methods, and counterfactual methods along five axes: whether the method gives attribute-level attribution; whether it produces an actionable counterfactual correction; whether it natively searches a mixed discrete-continuous space; whether it judges validity by a per-attribute reconstruction-error threshold; and whether it targets the accounting domain. Existing AE auditing methods detect and (in recent work) attribute anomalies but stop short of remediation; general counterfactual methods produce corrections but do not exploit per-attribute reconstruction error and handle mixed discrete-continuous data only by post-hoc rounding or manifold copying. The gap this work fills is the combination of all five: attribute-level attribution feeding a native mixed-space counterfactual search with per-attribute validity, for accounting journal entries.

III. PROBLEM SETTING AND DATA

A. DATASET

We use the SAP-format accounting journal entry dataset from Ref. [1].[2] The original dataset contains both normal and anomalous transactions, but the original anomalies lack ground-truth records of which attributes were modified. Since ground truth is essential for evaluating CF generation, we extracted only normal transactions (532,789 entries) and injected 120 anomalies across four subtypes with recorded modified attributes. The resulting dataset comprises 532,909 entries (532,789 normal + 120 injected anomalies). The data consists of eight attributes shown in Table 2.

Mutability specifies the degree to which each attribute can be modified during CF generation. Mutable attributes (DMBTR, WRBTR) can be freely changed; semi-mutable attributes (categorical) can only take values observed in the training data; the immutable attribute (BUKRS) cannot be changed at all. We treat BUKRS as immutable because companies do not rewrite the company code of an existing journal entry, so a BUKRS change is not a realistic remediation. We exclude BUKRS from anomaly injection for the same reason.

Preprocessing applies one-hot encoding for categorical attributes and log-transform followed by min-max normalization for numeric attributes, yielding a 221-dimensional feature vector.

The dataset differs from real-world journal data in three ways that matter here. (1) Debit and credit lines do not balance within a journal entry, so balance-violation anomalies are out of scope. (2) The number of distinct codes per categorical attribute is small: HKONT has three values against hundreds to thousands in real systems, KTOSL has only nine, and WAERS also has nine. (3) DMBTR is almost independent of the categorical attributes (Fig. 6). Section VI-G returns to what these gaps mean for CF performance.

[2]https://raw.githubusercontent.com/GitiHubi/deepAI/master/data/fraud_dataset_v2.csv

B. ANOMALY SUBTYPES

Four types of anomalies were injected for evaluation (Table 3). Global anomalies have individual attribute values that are outliers of the marginal distribution, while local anomalies have individually common values whose combination is unusual (outliers of the conditional distribution) [27].

C. PROBLEM FORMULATION

Given a transaction $\mathbf{x}$ flagged as anomalous, we generate a counterfactual $\mathbf{x}'$ satisfying two conditions:

1) **Validity**: $\mathbf{x}'$ must be classified as normal by the AE across all attributes:
$$\max_{\text{attr}} \frac{\text{MSE}_{\text{attr}}(\mathbf{x}')}{\theta_{\text{attr}}} \leq 1.0 \tag{1}$$
2) **Sparsity**: The number of attributes modified in $\mathbf{x}'$ should be minimized, so the auditor sees a short correction.

Here $\text{MSE}_{\text{attr}}(\mathbf{x}')$ is the mean reconstruction error over columns belonging to attribute *attr*, and θ_{attr} is the per-attribute threshold.

IV. PROPOSED METHOD

A. PIPELINE OVERVIEW

The proposed pipeline has five stages, shown in Fig. 1:

1) **Training**: Train the autoencoder on a curated set of normal transactions (Section IV-B).
2) **Detection**: Flag anomalies using a global BCE loss threshold (Section IV-D).
3) **RESHAPE**: For each flagged anomaly, compute attribute-level SHAP attributions, which will serve as the attribute search priority in stage 4 (Section IV-E).
4) **CF generation**: Run RESHAPE-guided beam search followed by gradient-based numeric refinement to produce a counterfactual (Sections IV-F–IV-G). Stages 3 and 4 are applicable to any flagged anomaly, but in this paper we run them only on the flagged true-positive injected anomalies, because evaluating exact match and partial hit requires ground-truth modified attributes (see Section V-A).
5) **Evaluation**: Check CF validity against per-attribute thresholds, and measure ground-truth alignment (exact match, partial hit) and sparsity.

Anomaly injection (Section III-B) is part of the experimental setup rather than the method itself; it supplies the ground-truth modified attributes that stage 5 needs to compute exact match and partial hit. The RESHAPE-guided CF generation in stages 3–4 is the main contribution of this paper.

TABLE 1. Positioning relative to representative prior work. ✓: supported; (✓): partially supported; —: not addressed.

Category	Method	Attribute-level attribution	Counterfactual remediation	Mixed discrete–continuous search	Per-attribute validity	Accounting domain
AE-based auditing	Schreyer et al. [1], [3]	—	—	—	—	✓
	Schultz & Tropmann-Frick [2]	—	—	—	—	✓
	Sattarov et al. [10]	✓	—	—	(✓)	✓
Attribution / RESHAPE	Antwarg et al. [23]	✓	—	—	—	—
	Mueller et al. (RESHAPE) [6]	✓	—	—	—	✓
	Torabi et al. [11]	✓	—	—	✓	—
Counterfactual	Wachter et al. [14]	—	✓	(✓)	—	—
	DiCE [15]	—	✓	(✓)	—	—
	FACE [16]	—	✓	(✓)	—	—
	Haldar et al. [21], Sulem et al. [22]	(✓)	✓	—	—	—
This work	RESHAPE-guided CF pipeline	✓	✓	✓	✓	✓

TABLE 2. Dataset attributes.

Attribute	Description	Type	Mutability
KTOSL	General ledger account key	Categorical	Semi-mutable
PRCTR	Profit center	Categorical	Semi-mutable
BSCHL	Posting key	Categorical	Semi-mutable
HKONT	General ledger account	Categorical	Semi-mutable
BUKRS	Company code	Categorical	Immutable
WAERS	Currency key	Categorical	Semi-mutable
DMBTR	Amount in local currency	Numeric	Mutable
WRBTR	Amount in document currency	Numeric	Mutable

TABLE 3. Anomaly subtype definitions and injection counts.

Category	Subtype	Modification	Count	Ground truth
Global	Unseen code	Replace one categorical attribute with a code unseen in training data	20	Modified attribute
Local	Unseen pair (KH)	Set KTOSL and HKONT each to common values, but the pair is unobserved	35	KTOSL, HKONT
	Unseen pair (HP)	Set HKONT and PRCTR each to common values, but the pair is unobserved	35	HKONT, PRCTR
	Unusual categ+amount	Change HKONT/KTOSL to a different common value and set DMBTR to p90–p99	30	Categorical attr + DMBTR

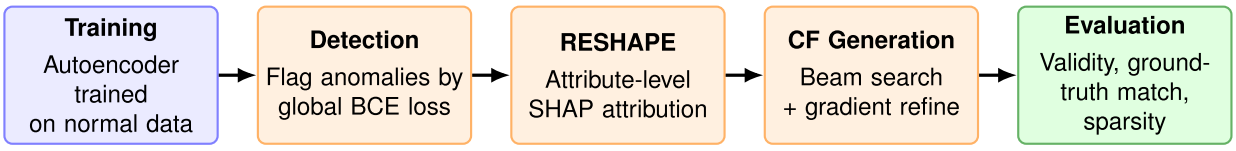


FIGURE 1. End-to-end pipeline. The autoencoder is trained on a curated set of normal transactions, global BCE loss flags anomalies, RESHAPE computes attribute-level SHAP attributions as the attribute search priority, beam search with gradient refinement then produces a counterfactual, and the counterfactuals are evaluated for validity, ground-truth alignment, and sparsity. Stages 3 and 4 are applicable to any flagged anomaly; in this paper, however, they are run only on the true-positive injected anomalies, since exact match and partial hit require ground-truth modified attributes (section V-A).

B. AUTOENCODER ARCHITECTURE

We use a bottleneck autoencoder with input dimension 221 and latent dimension 3. Both the encoder and decoder consist of fully connected layers with LeakyReLU activations, following the architecture of Schreyer et al. [1]. To prevent leakage among the data used for training, threshold selection, and evaluation (a concern when a single pool of normal entries serves all three roles), we partition the 532,789 normal transactions into disjoint training (70%, 372,952 entries), validation (15%, 79,918 entries), and test (15%, 79,919 entries) sets; the 120 injected anomalies are held out of all three partitions and used only as the evaluation target. This is a deliberate divergence from the original Schreyer et al. setup, which trains on the full unlabeled dataset because anomaly labels are unavailable in their scenario; in ours, the injected anomalies carry ground-truth modified

attributes that are reused for CF evaluation, so keeping them out of both training and threshold selection is necessary to avoid leakage between the evaluation target and the model. The autoencoder is trained on the training partition only, and the min–max normalization parameters for the numeric attributes are fit on the training partition alone and then applied unchanged to the validation and test partitions.

C. IMPLEMENTATION DETAILS

For reproducibility we list the full configuration. The encoder maps the 221-dimensional input through fully connected layers of sizes [512, 256, 128, 64, 32, 16, 8, 4, 3] down to the 3-dimensional latent code, and the decoder mirrors this with layers [4, 8, 16, 32, 64, 128, 256, 512, 221]. Every hidden layer uses a LeakyReLU activation with negative slope 0.4, weights are initialized with Xavier uniform initialization, and no dropout is applied. The network is trained to minimize binary cross-entropy with logits (`BCEWithLogitsLoss`) using two separate Adam optimizers (one for the encoder, one for the decoder), each with learning rate 10^{-3} and batch size 128, for 10 epochs with no early stopping. All randomness (weight initialization, mini-batch shuffling, and the train/validation/test split) is seeded with 1234; anomaly injection uses seed 42. The final training BCE loss is 0.0100. The remaining CF-generation hyperparameters are listed in Table 5.

D. TWO-STAGE THRESHOLDING

We use different thresholding criteria for anomaly detection and CF validity assessment. Detection uses a single global BCE loss threshold; CF validity is assessed per attribute.

1) STAGE 1: GLOBAL BCE LOSS FOR ANOMALY DETECTION

The detection threshold is derived from the clean training data BCE loss distribution:

$$\theta_{\mathrm{BCE}} = \mu_{\mathrm{BCE}} + k_{\mathrm{det}}\,\sigma_{\mathrm{BCE}} \tag{2}$$

where μ_{BCE} and σ_{BCE} are the mean and standard deviation of the AE's BCE reconstruction loss over the clean transactions in the *training* partition ($\mu_{\mathrm{BCE}} = 0.0100$, $\sigma_{\mathrm{BCE}} = 0.0079$). We select k_{det} *without using any anomaly labels*: we sweep k_{det}, measure the false-positive rate (FPR) on the held-out clean *validation* partition for each value, and choose the smallest k_{det} whose validation FPR falls at or below a target of 5×10^{-4}. This yields $k^*_{\mathrm{det}} = 9.5$ and $\theta_{\mathrm{BCE}} = 0.0847$. Because the validation partition contains only clean transactions, the injected anomalies never enter threshold selection; recall is measured afterwards on the held-out anomalies, and false positives on the clean *test* partition (Section V-B).

2) STAGE 2: PER-ATTRIBUTE THRESHOLDS FOR CF VALIDITY

A counterfactual $\mathbf{x}'$ is valid only if the per-attribute reconstruction error is below threshold for every attribute. Per-attribute thresholding prevents anomalies in one attribute from being masked by normal reconstruction in others.

The per-attribute reconstruction error is defined as:

$$\mathrm{MSE}_{\mathrm{attr}}(\mathbf{x}) = \frac{1}{n_{\mathrm{attr}}} \sum_{j \in \mathcal{C}_{\mathrm{attr}}} \left(\sigma(\hat{\mathbf{x}})_j - \mathbf{x}_j\right)^2 \tag{3}$$

where $\mathcal{C}_{\mathrm{attr}}$ is the set of columns belonging to attribute *attr*, $n_{\mathrm{attr}} = |\mathcal{C}_{\mathrm{attr}}|$, $\hat{\mathbf{x}} = \mathrm{decoder}(\mathrm{encoder}(\mathbf{x}))$, and σ is the sigmoid function.

Both the AE training and the threshold computation (global BCE and per-attribute) use only clean (non-anomalous) transactions from the training partition, which we assume to be available as a curated reference subset. This places the method within the semi-supervised anomaly detection framework, where the system has access to known-normal data but no labeled anomalies. Per-attribute thresholds are defined as:

$$\theta_{\mathrm{attr}} = \mu_{\mathrm{attr}} + k_{\mathrm{cf}}\,\sigma_{\mathrm{attr}} \tag{4}$$

where μ_{attr} and σ_{attr} are the mean and standard deviation of $\mathrm{MSE}_{\mathrm{attr}}$ over the clean training data, with $k_{\mathrm{cf}} = 10$. We deliberately keep k_{cf} comparable to k_{det}: a stricter per-attribute threshold forces the beam search to identify all ground-truth anomalous attributes rather than stopping early after fixing only one, while the detection threshold (k^*_{det}=9.5) keeps the false-positive count low.

The violation ratio is defined as:

$$v_{\mathrm{attr}}(\mathbf{x}) = \frac{\mathrm{MSE}_{\mathrm{attr}}(\mathbf{x})}{\theta_{\mathrm{attr}}} \tag{5}$$

A CF is invalid if any attribute exceeds $v_{\mathrm{attr}} > 1.0$.

Table 4 shows the per-attribute thresholds.

TABLE 4. Per-attribute thresholds ($k_{\mathrm{cf}} = 10$), computed on the clean training partition and used for CF validity assessment.

Attribute	Threshold θ_{attr}
KTOSL	0.065
PRCTR	0.047
BSCHL	0.003
HKONT	0.031
BUKRS	0.006
WAERS	0.005
DMBTR	0.152
WRBTR	0.994

3) WHY TWO STAGES

The global BCE threshold aggregates all features into a single scalar, providing a simple detection criterion. Per-attribute thresholding for CF validity checks that every attribute individually falls within the normal range. Since the overall reconstruction error is a column-count-weighted average of per-attribute MSEs, anomalies in low-dimensional attributes (e.g., DMBTR with 1 column) can be masked by normal reconstruction in high-dimensional attributes (e.g., KTOSL with 9 columns). Per-attribute assessment avoids this masking problem.

E. RESHAPE (SHAP ATTRIBUTION)

RESHAPE (Reconstruction Error SHapley Additive exPlanations) [6] applies SHAP's KernelExplainer to the AE reconstruction loss to quantify each attribute's contribution to the anomaly.

1) Define the prediction function $f(\mathbf{x}) = \mathrm{BCELoss}(\hat{\mathbf{x}}, \mathbf{x})$ as a scalar output.
2) Compute feature-level SHAP values using KernelExplainer (100 background samples, 500 perturbations).
3) Aggregate one-hot column SHAP values to the original attribute level by summation.

The attribute with the highest SHAP value contributes most to the reconstruction loss and is interpreted as the primary anomaly cause. These attribution values determine the attribute search priority in CF generation.

F. BEAM SEARCH FOR CF GENERATION

The proposed CF generation algorithm is shown in Algorithm 1.

Algorithm 1 RESHAPE-Guided Beam Search for Counterfactual Generation

Require: Anomalous transaction $\mathbf{x}$, RESHAPE priority π, beam width W, max steps T
Ensure: Counterfactual $\mathbf{x}^*$

```
beam ← [x], x*_valid ← null
for t = 1 to T do
    candidates ← []
    for each candidate c ∈ beam do
        for each attribute a ∈ π (priority order) do
            for each candidate value v do
                c' ← c, c'_a ← v
                Append (Obj(x, c'), c') to candidates
            end for
        end for
        Also explore pairwise categorical mutations
    end for
    Deduplicate, sort by Obj ascending, keep top W
    if valid CF found then
        x*_valid ← CF with fewest changes
        break
    end if
end for
if x*_valid ≠ null then
    Apply gradient-based numeric optimization (§IV-G)
end if
return x*_valid (or beam top if none found)
```

At each step of Algorithm 1, the objective function Obj is evaluated for all candidates, and only the top W are retained. This greedy selection heuristically favors low-Obj candidates. The objective function is:

$$\mathrm{Obj}(\mathbf{x}, \mathbf{x}') = \underbrace{\max(0, v_{\max}(\mathbf{x}') - 1.0)}_{\text{validity loss}} + \lambda_{\text{sparse}} \cdot s(\mathbf{x}, \mathbf{x}') \quad (6)$$

where $v_{\max}(\mathbf{x}') = \max_{\text{attr}} v_{\text{attr}}(\mathbf{x}')$ is the maximum per-attribute violation ratio and $s(\mathbf{x}, \mathbf{x}') = |\{\text{attr} : x_{\text{attr}} \neq x'_{\text{attr}}\}|$ is the number of modified attributes.

The sparsity term counts changed attributes rather than using a continuous distance. Numeric magnitudes such as DMBTR would dominate a distance-based term, and beam search would then cling to the candidates closest to the original numeric values and never try genuinely different amounts. We defer sample-specific refinement of numeric values to the post-hoc gradient descent step (Section IV-G).

Hyperparameters are listed in Table 5.

G. GRADIENT-BASED NUMERIC OPTIMIZATION

After beam search finds a valid CF, we refine the numeric attributes via gradient descent to further reduce reconstruction error.

The following loss is minimized over normalized numeric attribute values:

$$\mathcal{L}_{\text{num}} = \sum_{a \in \mathcal{A}_{\text{num}}} \left(\sigma(\hat{\mathbf{x}})_a - \mathbf{x}'_a\right)^2 \quad (7)$$

where $\mathcal{A}_{\text{num}}$ is the set of changed numeric attributes, $\mathbf{x}'_a$ is the CF candidate value for attribute a, and $\sigma(\hat{\mathbf{x}})_a$ is the AE output after sigmoid activation. Adam optimizer is used with 100 steps and learning rate 0.05. Values are clamped to the [0, 1] normalized range.

Beam search is limited to 20 discrete candidate values for numeric attributes, so different samples tend to receive the same values. Gradient optimization searches the continuous space instead, finding optima specific to each sample's categorical configuration.

V. EXPERIMENTS

A. EXPERIMENTAL SETUP

The AE was trained on the 372,952-entry training partition of clean transactions only (Section IV-C), for 10 epochs (final training BCE loss 0.0100). The detection threshold $k^*_{\text{det}} = 9.5$ was selected on the 79,918-entry clean validation partition by the label-free FPR criterion of Section IV-D, without using any anomaly labels. Detection was then evaluated on the 79,919-entry clean test partition together with the 120 held-out injected anomalies: the detector flagged all 120 anomalies (100% recall) and 31 clean-test entries as false positives (FPR 3.9×10^{-4}). RESHAPE attribution and CF generation were subsequently run only on the 120 true-positive injected anomalies, because evaluating exact match and partial hit requires ground-truth modified attributes, which are available only for injected samples. The CF generator itself is applicable to any flagged anomaly, and we expect it to produce useful CFs for the false positives as well, but those cannot be scored against ground truth and are analyzed separately in Section V-H. Per-attribute thresholds for the beam search's internal validity check use k_{cf}=10 (Table 4). Evaluation metrics include detection rate (recall), exact match (CF changed attributes exactly equal the injected ground-truth attributes), partial hit (at least one overlapping

TABLE 5. Hyperparameters.

Parameter	Value	Role
beam_width (W)	10	Beam width
max_steps (T)	8	Maximum search steps
λ_{sparse}	0.8	Sparsity weight
k_{det}	9.5	Detection σ multiplier
k_{cf}	10	CF validity σ multiplier
SHAP background samples	100	KernelExplainer
SHAP perturbations	500	KernelExplainer
Categorical candidates	top-10 (single), top-5 (pair)	Mutation candidates
Numeric candidates	logspace(p1, p99, 20)	20 log-spaced values
Gradient optimization steps	100	Post-hoc refinement
Gradient optimization learning rate	0.05	Adam optimizer

attribute), and mean sparsity (number of changed attributes). We also report the beam search convergence rate, i.e., the fraction of cases where beam search produced a CF that satisfies all per-attribute thresholds within the maximum of T search steps.

B. DETECTION PERFORMANCE

Table 6 presents detection and CF generation performance by anomaly subtype. At the label-free operating point $k^*_{\text{det}} = 9.5$ ($\theta_{\text{BCE}} = \mu_{\text{BCE}} + 9.5\,\sigma_{\text{BCE}} = 0.0847$), the overall detection rate is 100% (120/120), with 31 false positives on the clean test partition (FPR 3.9×10^{-4}). All four subtypes are detected at 100% recall.

Table 7 shows how detection responds to the detection threshold parameter k_{det}, reporting recall on the 120 held-out anomalies and the false-positive rate separately on the clean validation and clean test partitions. Recall stays at 100% for every $k_{\text{det}} \leq 9.5$ and only begins to drop at $k_{\text{det}} \geq 9.75$ (one anomaly missed), reaching 98.3% by $k_{\text{det}} = 13.5$ (where the clean-test FPR first reaches zero) and 97.5% at $k_{\text{det}} = 20$. The validation FPR crosses the 5×10^{-4} target between $k_{\text{det}} = 9.25$ (5.8×10^{-4}) and $k_{\text{det}} = 9.5$ (3.1×10^{-4}), which is what fixes $k^*_{\text{det}} = 9.5$. This label-free choice happens to coincide with the largest k_{det} still at 100% recall, but it is reached *without* consulting the anomaly labels: the validation and test FPRs track each other closely (3.1×10^{-4} vs. 3.9×10^{-4} at k^*_{det}), confirming that a threshold tuned on the validation partition generalizes to the test partition. In practice, the remaining false positives can be prioritized for investigation by ranking them by BCE loss magnitude (Section V-H).

C. CF GENERATION PERFORMANCE

The CF generator exactly matched the ground-truth modified attributes in 30 of 120 true-positive cases (25.0%, exact match) and hit at least one ground-truth attribute in 99 of 120 (82.5%, partial hit). Mean sparsity is approximately 1.18, so auditors typically need to examine only 1–2 attributes. Beam search converged to a CF that satisfies all per-attribute thresholds in all 120 of 120 cases within the maximum of T=8 search steps. We treat this convergence rate as an internal diagnostic rather than a quality metric for the CF itself, since beam search is optimizing directly for that criterion (Section IV-F).

Exact match and partial hit vary substantially across subtypes. Unseen code has 100% exact match because the anomaly involves a single attribute modification and the CF identifies the same attribute. Unseen pair (KH) has 100% partial hit with 11% exact match; in most cases, changing one of the two attributes (typically KTOSL) is enough to bring every per-attribute violation ratio below 1, and only in 11% of cases does beam search modify both ground-truth attributes. Unseen pair (HP) shows 74% partial hit and 17% exact match (Fig. 3); the lower partial-hit rate compared to KH is because the AE's per-attribute thresholds for HKONT and PRCTR can also be cleared by changing a third correlated attribute, and beam search often takes those "sideways" paths and misses the ground truth (see Section VI-B). Unusual categ+amount has 60% partial hit but 0% exact match; this is discussed in Section VI-C.

Fig. 2 shows the max violation ratio before and after CF application. Violation drops across all subtypes.

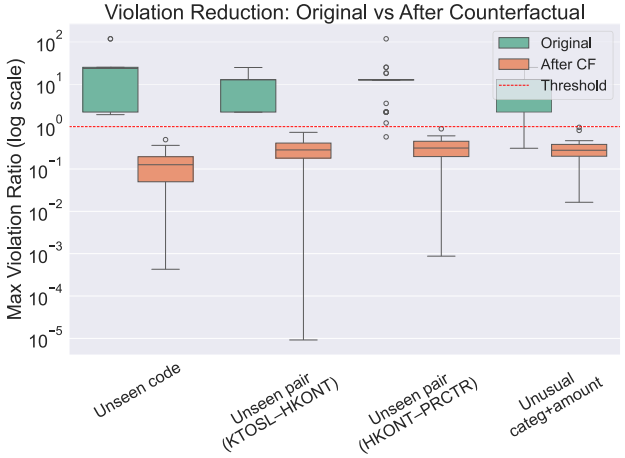


FIGURE 2. Max violation ratio before (green) and after (orange) CF application, on a log scale. Red dashed line: validity threshold (violation ratio = 1.0). Outliers near ∼320 reflect MSE saturation on low-cardinality attributes.

Fig. 3 shows a heatmap of CF changed attributes and ground-truth attributes for all 120 true-positive injected anomalies. Unseen code shows strong alignment between CF

TABLE 6. Detection and CF generation performance by anomaly subtype. Detection (recall) uses the global BCE loss threshold at the label-free operating point k^*_{det}=9.5. Exact match and partial hit compare the CF's changed attributes against the injected ground-truth attributes.

Subtype	N	Detection	Exact match	Partial hit	Sparsity
Unseen code	20	20/20 (100%)	20/20 (100%)	20/20 (100%)	1.0
Unseen pair (KH)	35	35/35 (100%)	4/35 (11%)	35/35 (100%)	1.3
Unseen pair (HP)	35	35/35 (100%)	6/35 (17%)	26/35 (74%)	1.3
Unusual categ+amt	30	30/30 (100%)	0/30 (0%)	18/30 (60%)	1.0
Overall	**120**	**120/120 (100%)**	**30/120 (25.0%)**	**99/120 (82.5%)**	**∼1.18**

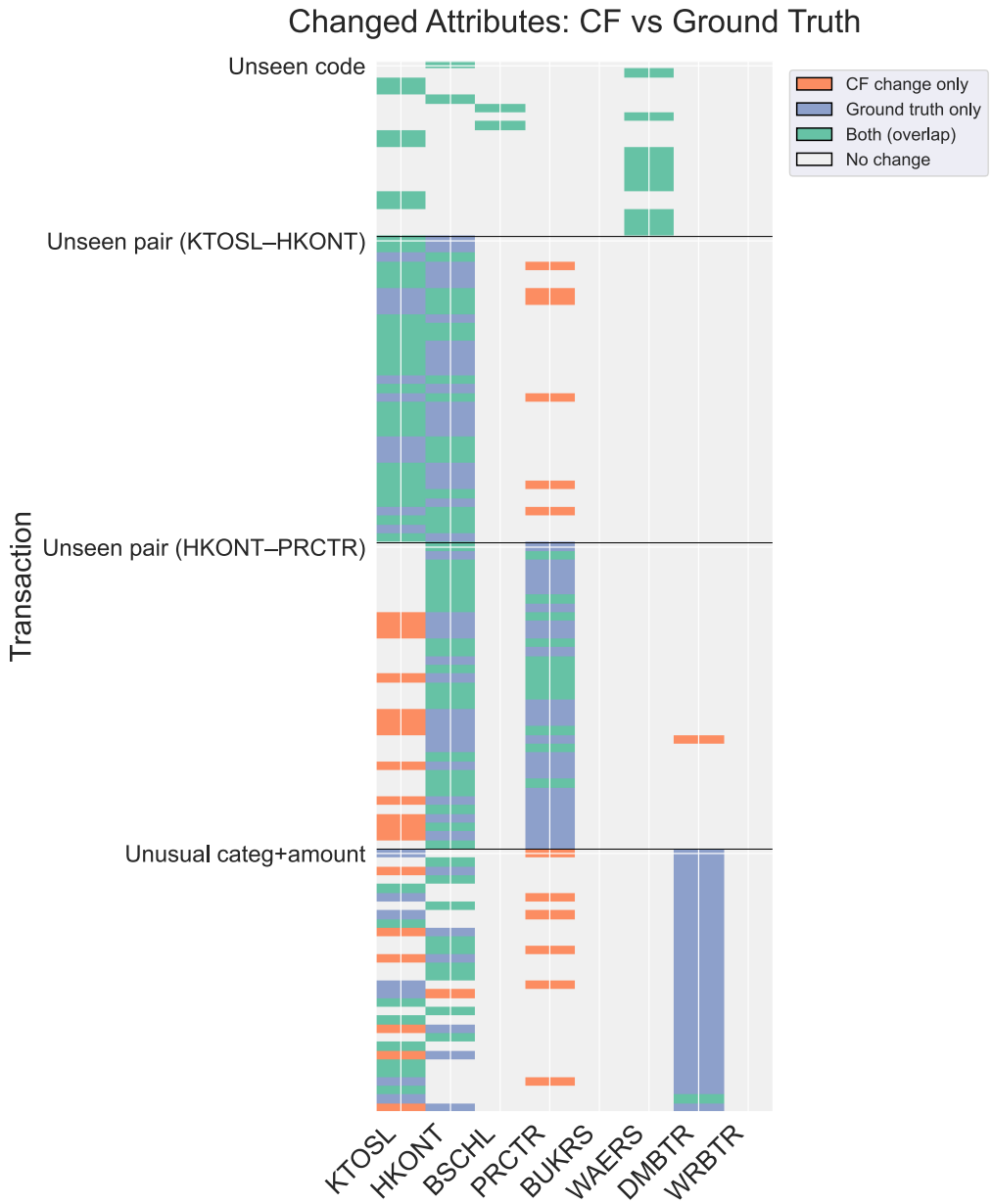


FIGURE 3. Heatmap of CF changed attributes (orange), ground-truth attributes (blue), and overlapping attributes (green). Grouped by anomaly subtype.

changes and ground truth. For unseen pairs, the CF often modifies only one of the two ground-truth attributes, which is sufficient to bring every per-attribute violation ratio below 1 (discussed in Section VI-B).

TABLE 7. Detection sensitivity to the detection threshold parameter k_{det}, with $\theta_{\text{BCE}} = \mu_{\text{BCE}} + k_{\text{det}}\sigma_{\text{BCE}}$. Recall is on the 120 held-out injected anomalies; FPR is reported separately on the clean validation and clean test partitions. The selected label-free operating point is k^*_{det}=9.5.

k_{det}	θ_{BCE}	Recall	FPR (val)	FPR (test)
6.0	0.0572	100.0%	2.1×10^{-3}	1.9×10^{-3}
7.0	0.0650	100.0%	1.2×10^{-3}	1.0×10^{-3}
8.0	0.0729	100.0%	1.1×10^{-3}	8.6×10^{-4}
9.0	0.0807	100.0%	5.9×10^{-4}	5.4×10^{-4}
9.25	0.0827	100.0%	5.8×10^{-4}	4.9×10^{-4}
9.5	**0.0847**	**100.0%**	$\mathbf{3.1 \times 10^{-4}}$	$\mathbf{3.9 \times 10^{-4}}$
9.75	0.0866	99.2%	2.8×10^{-4}	3.1×10^{-4}
10.0	0.0886	99.2%	2.3×10^{-4}	2.5×10^{-4}
11.0	0.0965	99.2%	1.3×10^{-4}	1.6×10^{-4}
12.0	0.1043	99.2%	3.8×10^{-5}	7.5×10^{-5}
13.5	0.1161	98.3%	0	0
15.0	0.1279	98.3%	0	0
20.0	0.1672	97.5%	0	0

D. ABLATION STUDY

To isolate the contribution of each design choice, we ablate three components on the 120 anomalies: (i) RESHAPE-guided search priority versus an unguided fixed attribute order, (ii) the gradient-based numeric optimization step on versus off, and (iii) the two-stage *per-attribute* CF validity criterion versus a single *global* BCE validity criterion. Table 8 reports the full $2 \times 2 \times 2$ factorial.

Three findings emerge. First, RESHAPE prioritization improves exact match by 7.5 percentage points under the per-attribute criterion (25.0% vs. 17.5%) and by 2.5 points under the global criterion, while leaving partial hit and sparsity essentially unchanged; ordering the search by attribution helps the beam reach the true causal attribute sooner. Second, the gradient step does not change exact match, partial hit, or sparsity on this dataset. This is expected: the gradient step refines a numeric attribute only when the beam has already selected it as a change, which almost never happens here because DMBTR is nearly independent of the categorical attributes (Section VI-C); we validate the module directly in Section V-G. Third, the single global criterion attains a higher exact match (30.0%) but at the cost of less sparse corrections (sparsity 1.34 vs. 1.18) and, in two configurations, counterfactuals that the per-attribute check rejects (validity 99.2%): the global criterion lets the search stop while a low-dimensional attribute still violates its own threshold, which is exactly the masking the per-attribute stage is designed to prevent (Section VI-A). The two-stage design therefore trades a small amount of exact match for sparser, per-attribute-valid corrections.

E. COMPARISON WITH COUNTERFACTUAL BASELINES

We compare against three mainstream CF methods on the same 120 anomalies under the same mutability constraints (BUKRS immutable; categorical attributes restricted to values observed in training; DMBTR and WRBTR free). To make classifier-oriented baselines applicable to our AE, we wrap the AE as a binary classifier $p_{\text{normal}}(\mathbf{x}) = \sigma(\alpha(1 - v_{\max}(\mathbf{x})))$ with $\alpha = 4$, so that an entry is "normal" exactly when its maximum per-attribute violation ratio $v_{\max}$ is below 1. *Wachter* [14] minimizes the classifier loss plus an L_1 distance by gradient descent in feature space, then projects back to the discrete code space; *DiCE* [15] samples diverse CFs against the same wrapped classifier; *FACE* [16] returns the nearest density-feasible clean training row. Table 9 reports the results, with the same per-attribute validity definition (Eq. 1) scoring every method.

Our method is the strongest overall on this data. It attains the highest exact match (25.0% vs. 18.3%/10.8%/6.7%), full per-attribute validity (100%), the lowest sparsity among the methods that achieve full validity (1.18 changed attributes vs. DiCE's 1.89 and FACE's 2.98), and substantially lower cost than Wachter (9.92 s) and DiCE (8.02 s) at 0.06 s per CF; FACE is faster still (0.004 s) but only because it copies a whole clean training row, at the cost of exactness (6.7%) and sparsity (2.98). Each baseline has a specific, interpretable weakness. *Wachter* produces a per-attribute-valid CF in only 33% of cases: its continuous gradient step, followed by an argmax projection onto discrete codes, repeatedly lands on the same saturated reconstruction error for code-substitution anomalies, so the projected CF often fails the per-attribute check even though a sparse valid CF provably exists (we return to this in Section VI-F); its nominally low sparsity (0.58) reflects that it frequently changes no attribute at all and fails validity, rather than producing tighter corrections. *FACE* is always valid and always partial-hit, but because it copies a whole clean row it changes nearly three attributes on average (sparsity 2.98) and rarely matches the ground-truth attribute set exactly (6.7%). *DiCE* reaches a slightly higher partial hit (86.7%) than ours but a much lower exact match (10.8%), with a less sparse and roughly 130× slower CF. The hybrid discrete-plus-gradient design with a per-attribute validity target is what produces accurate, sparse, and valid corrections at low cost on this mixed discrete-continuous data.

TABLE 8. Ablation over RESHAPE-guided priority, the gradient step, and the validity criterion (per-attribute vs. single global BCE), on the 120 anomalies. The first row is the proposed full configuration.

RESHAPE	Grad.	Validity	Exact	Partial	Spars.	Valid
✓	✓	per-attr	**25.0%**	**82.5%**	**1.18**	**100%**
✓	✓	global	30.0%	97.5%	1.34	99.2%
✓	—	per-attr	25.0%	82.5%	1.18	100%
✓	—	global	30.0%	97.5%	1.34	100%
—	✓	per-attr	17.5%	80.0%	1.18	100%
—	✓	global	27.5%	96.7%	1.34	99.2%
—	—	per-attr	17.5%	80.0%	1.18	100%
—	—	global	27.5%	96.7%	1.34	100%

F. HYPERPARAMETER SENSITIVITY

We swept the beam width $W \in \{5, 10, 20\}$, the maximum search steps $T \in \{4, 8, 12\}$, and the sparsity weight $\lambda_{\text{sparse}} \in \{0.4, 0.8, 1.6\}$ (the full $3 \times 3 \times 3$ grid; Table 10 shows a one-at-a-time slice around the default W=10, T=8, λ_{sparse}=0.8). Exact match, partial hit, and sparsity are insensitive to W

TABLE 9. Counterfactual baseline comparison on the 120 anomalies. Validity uses the per-attribute criterion (Eq. 1). Time is wall-clock seconds per CF.

Method	Exact	Partial	Valid	Spars.	Time (s)
Ours	**25.0%**	82.5%	**100%**	1.18	0.06
Wachter [14]	18.3%	35.8%	33.3%	0.58†	9.92
DiCE [15]	10.8%	86.7%	100%	1.89	8.02
FACE [16]	6.7%	100%	100%	2.98	0.004

†Wachter's low sparsity reflects frequent empty CFs that fail the validity check, not tighter corrections.

and T across the tested ranges: with $\lambda_{\text{sparse}} \leq 0.8$, every configuration yields exact match 25.0%, partial hit 82.5%, sparsity 1.18, and full validity. The reason is that RESHAPE prioritization places the causal attributes near the front of the search, so even the narrowest beam (W=5) and the shortest horizon (T=4) already find the same corrections; larger W and T only add runtime. The single influential parameter is λ_{sparse}: raising it to 1.6 over-penalizes changes, dropping exact match to 21.7%, partial hit to 78.3%, and validity to 94.2% as the search sometimes refuses a needed second change. For deployment we therefore recommend the smallest W and T that fit the latency budget and a moderate $\lambda_{\text{sparse}} \leq 0.8$; we use W=10, T=8, λ_{sparse}=0.8.

TABLE 10. Hyperparameter sensitivity (one parameter varied at a time around the default, shown in bold). Exact/partial/sparsity are unchanged by W and T and degrade only for large λ_{sparse}.

W	T	λ_{sparse}	Exact	Partial	Spars.	Valid
10	**8**	**0.8**	**25.0%**	**82.5%**	**1.18**	**100%**
5	8	0.8	25.0%	82.5%	1.18	100%
20	8	0.8	25.0%	82.5%	1.18	100%
10	4	0.8	25.0%	82.5%	1.18	100%
10	12	0.8	25.0%	82.5%	1.18	100%
10	8	0.4	25.0%	82.5%	1.18	100%
10	8	1.6	21.7%	78.3%	1.08	94.2%

G. EFFECTIVENESS OF GRADIENT-BASED NUMERIC OPTIMIZATION

Because the unusual categ+amount subtype reaches 0% exact match (the CF never corrects DMBTR), it is fair to ask whether the gradient-based numeric module is actually effective. The 0% is a property of the dataset, not of the module: in this data DMBTR is nearly independent of the categorical attributes (Section VI-C), so the amount carries no detectable signal to correct. To test the module in isolation, we constructed a controlled probe. Starting from clean test entries, we corrupted a single numeric attribute (DMBTR or WRBTR) by a large multiplicative factor so that its per-attribute reconstruction error is driven far above threshold, and then ran the gradient step alone with the ground-truth attribute set to that one numeric attribute. Across 153 probes (100 DMBTR, 53 WRBTR), the gradient step returned the per-attribute error below threshold in all 153 cases, reducing it by a median of essentially 100% to a residual that is effectively zero (Table 11, Fig. 4). The module thus reliably normalizes a genuinely anomalous numeric attribute when one exists; it stays silent on the injected unusual categ+amount anomalies only because those amounts are not anomalous given the categorical context in this dataset.

TABLE 11. Gradient module validation on controlled single-numeric-attribute probes. e denotes the per-attribute reconstruction error; "recovered" means e ended below the attribute's validity threshold.

Attr.	Probes	Recovered	Med. e before	Med. e after
DMBTR	100	100	0.185	8.4×10^{-7}
WRBTR	53	53	1.149	1.3×10^{-5}

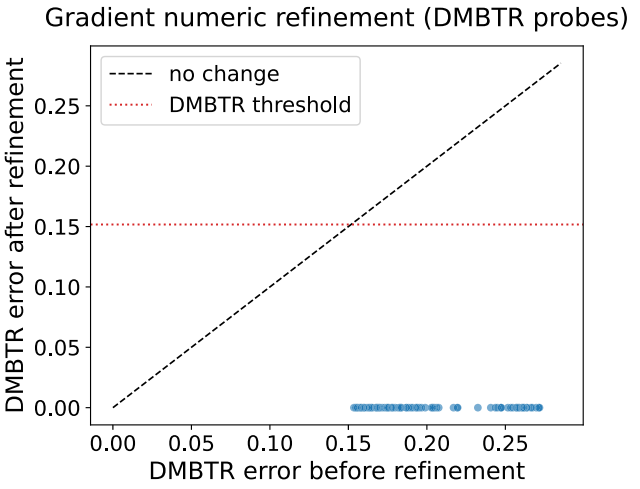


FIGURE 4. Per-attribute reconstruction error of the corrupted DMBTR attribute before (x-axis) versus after (y-axis) the gradient refinement step, over the 100 DMBTR probes. The dotted line marks the DMBTR validity threshold and the dashed diagonal marks "no change"; every probe ends far below the threshold, confirming that the gradient module normalizes a genuinely anomalous numeric attribute.

H. FALSE-POSITIVE ANALYSIS

We re-derived the false-positive set at $k^*_{\text{det}} = 9.5$ on the clean test partition (31 entries, FPR 3.9×10^{-4}), and characterized each by which attribute drives its BCE loss, its RESHAPE attribution, and whether it carries a rare categorical code. The false positives are homogeneous: in 26 of 31 the dominant per-attribute error is KTOSL (the remaining five split as four PRCTR and one BSCHL), and nearly all of them carry a relatively infrequent categorical code, most often the company code (BUKRS), appearing in roughly 0.5% of training rows. None of the 31 carries a code rarer than 0.1%, so these are not unseen codes but legitimate, lightly populated company×account combinations that the low-capacity (latent-3) AE reconstructs slightly worse than average. Fig. 5 shows the per-attribute error profile of the 31 false positives.

Table 12 quantifies three label-free reduction strategies against their recall cost. Ranking by BCE loss and triaging

the entries that fall below the *lowest* true-positive BCE deprioritizes 6 of the 31 at no recall cost, a free improvement to the auditor's worklist order. A per-attribute veto (flag only if at least one attribute also exceeds its k_{cf} threshold) removes 4 of the 31 but costs two true positives (98.3% recall), and a stricter two-of-N veto removes 30 of the 31 but is far too aggressive, cutting recall to 67.5%. In a real workflow we would therefore combine the free BCE-rank triage with an optional per-attribute veto when the small recall cost is acceptable; the broader point is that the false positives are concentrated and rankable, not arbitrary.

TABLE 12. False-positive reduction strategies at k^*_{det}=9.5 (31 false positives, 120 anomalies). Recall is over the 120 injected anomalies; the BCE-rank triage deprioritizes rather than discards, so recall is unaffected.

Strategy	FP removed	FP left	Recall
BCE-rank triage*	6	25	100%
Per-attr. veto (≥ 1 over k_{cf})	4	27	98.3%
Two-of-N veto (≥ 2 over k_{cf})	30	1	67.5%

*Triages entries below the lowest true-positive BCE; deprioritized in the worklist, not removed.

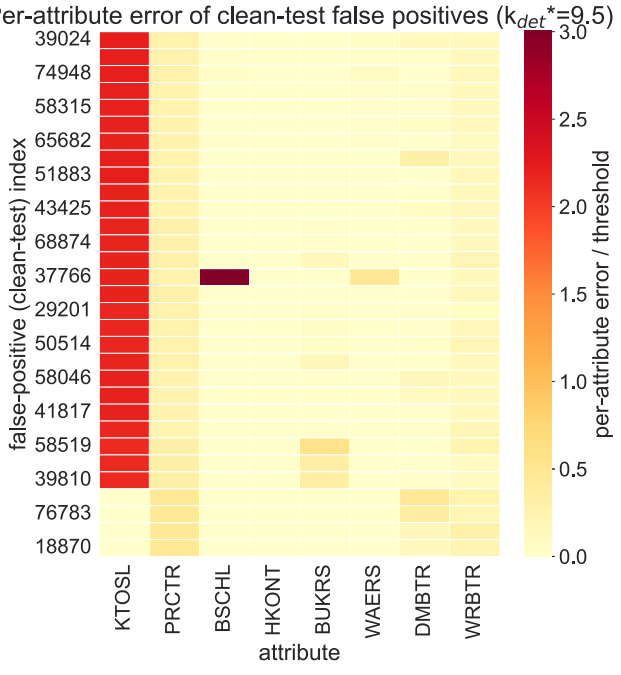


FIGURE 5. Per-attribute reconstruction error relative to each attribute's threshold for the 31 clean-test false positives (rows, sorted by BCE), at k^*_{det}=9.5. The error is concentrated in KTOSL for most false positives, while BUKRS stays near its threshold despite carrying an infrequent company code.

VI. DISCUSSION

A. BENEFITS OF TWO-STAGE THRESHOLDING

The two-stage design uses global BCE loss for detection and per-attribute thresholds for CF validity. Global BCE thresholding is simple: it aggregates all features into a single scalar. Per-attribute thresholding holds each attribute to its own normal range. These thresholds vary widely (BSCHL: 0.003 vs. WRBTR: 0.994; Table 4); a single aggregate score cannot capture that much granularity, and the ablation in Section V-D confirms that replacing the per-attribute check with a single global criterion admits counterfactuals that leave a residual per-attribute violation.

B. ANALYSIS OF PARTIAL HIT AND AE RECONSTRUCTION CHARACTERISTICS

Partial hit is 82.5% while exact match is 25.0%. This gap arises because the AE's reconstruction error does not necessarily increase for all ground-truth attributes.

For unseen pair anomalies, the ground truth involves two attributes (e.g., KTOSL and HKONT), but changing only one may suffice to bring every per-attribute violation ratio below 1. For KTOSL–HKONT pairs, the CF changes only KTOSL in most cases (exact match 11%, partial hit 100%). This occurs because the AE has learned enough co-occurrence structure that normalizing one attribute is sufficient to satisfy the per-attribute thresholds for both.

Unusual categ+amount has 60% partial hit, meaning CFs identify at least one correct attribute in the majority of cases. However, exact match is 0% because, as discussed in Section VI-C, the CF does not modify DMBTR since the AE does not detect amount anomalies in this dataset.

Beam search can reach validity without touching the injected attributes. In 21 of the 120 true-positive cases, the converged CF shares no attribute with the ground truth at all (12 of 30 unusual categ+amount and 9 of 35 HP cases). The beam search optimizes validity loss + $\lambda_{\mathrm{sparse}}\, s(\mathbf{x}, \mathbf{x}')$ rather than any ground-truth overlap, so if the AE has learned correlations between attributes, changing an uninjected attribute can drop the injected attribute's per-attribute reconstruction error by moving the input into a neighbourhood the AE considers normal, and the search will take that "sideways" path when it yields a lower objective. In effect, converging to a CF that satisfies all per-attribute thresholds is a necessary but not sufficient condition for explaining the true anomaly; partial hit is the stricter alignment metric.

C. LIMITATIONS OF UNUSUAL CATEG+AMOUNT DETECTION

The 0% exact match for the unusual categ+amount subtype results from the absence of conditional dependence between categorical attributes and amounts in this dataset. Fig. 6 shows that the DMBTR distribution is nearly identical regardless of HKONT code, so the AE does not learn a "this HKONT code implies this amount range" structure, and the CF proposes only categorical corrections. This looks like a property of the dataset, not a limitation of the AE itself; datasets with genuine categorical-amount dependencies may yield different results. Section V-G confirms this reading directly: in a controlled probe where a numeric attribute is the true anomaly, the gradient module drives its per-attribute error to essentially zero in all 153 cases, so the

0% exact match here reflects the absent categorical×amount dependence rather than an ineffective numeric module.

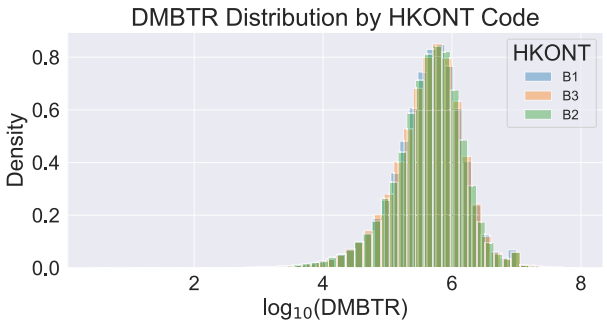


FIGURE 6. DMBTR distribution by HKONT code, computed over the 532,789 clean (non-anomalous) transactions across all three HKONT codes (B1, B2, B3) in the dataset. The distribution is nearly identical regardless of code, confirming the absence of categorical×amount conditional dependence in this dataset.

D. COMPLEMENTARITY OF RESHAPE AND CF

RESHAPE and CF answer different questions about the same anomaly (Table 13).

RESHAPE directs auditors to the problematic attributes; the CF then shows what a corrected entry would look like. Neither alone is sufficient: cause ranking without remediation leaves the auditor to guess, and remediation without cause ranking has no justification to cite.

E. RESIDUAL RISKS AND LIMITATIONS

At the label-free operating point $k^*_{\text{det}} = 9.5$, all 120/120 anomalies are detected with 31 false positives on the clean test partition. Section V-H characterizes those false positives and shows they are concentrated and rankable; lowering k_{det} only adds false positives (recall is already saturated), while raising it past 9.5 starts to drop true positives (Table 7).

This evaluation covers only four injected anomaly patterns. We have not tested against real-world anomalies (e.g., subtle amount manipulations or multi-attribute fraud), which are likely harder to detect than injected ones, and the dataset's low categorical cardinality and absent categorical×amount dependence (Section VI-C) shape several of our results.

The 31 false positives at $k^*_{\text{det}} = 9.5$ cost auditor review time. BCE loss ranking lets the auditor triage them at no recall cost (Section V-H), and adjusting k_{det} (Table 7) controls the recall–precision trade-off.

F. WHY THE BASELINES FALL SHORT

Section V-E and Table 9 quantify the gap to Wachter, DiCE, and FACE; here we explain it. The recurring failure of the gradient baseline on this data is a coupling effect: the per-attribute reconstruction errors are *not* independent. When a categorical code is substituted, fixing the causal attribute returns the latent code to the normal manifold and drops every attribute's error at once, so the validity surface is sharply non-convex in the continuous relaxation. Wachter's gradient descent slides along this relaxation and then rounds to the nearest discrete code, which on code-substitution anomalies repeatedly lands back on the same saturated error, hence its 33% validity, even though a one-attribute valid CF demonstrably exists (our beam search finds it). FACE sidesteps the discrete-continuous problem by copying a whole clean neighbour, which guarantees validity but sacrifices sparsity and exactness. The RESHAPE-guided beam search instead searches the discrete code space directly, with the gradient step reserved for genuinely numeric corrections, which is why it alone returns sparse, per-attribute-valid corrections on this mixed data.

The RESHAPE prioritization is also what keeps the search tractable: without it, the 221-dimensional mixed space would require near-exhaustive enumeration. Extending from 2-attribute to 3-attribute simultaneous mutations increases candidates from $\binom{6}{2} \times 5^2 = 375$ to $\binom{6}{3} \times 5^3 = 2{,}500$ ($\sim$6.7×), and 4-attribute mutations reach $\binom{6}{4} \times 5^4 = 9{,}375$; for datasets with many attributes this combinatorial growth would require more aggressive pruning or a learned search policy.

G. PRACTICAL APPLICABILITY TO AUDITING

In a typical audit workflow, the AE would first flag anomaly candidates from a large transaction set. For each flagged entry, RESHAPE ranks the attributes most likely responsible, and the CF generator proposes a minimal correction—on average about 1.18 attribute changes, so the auditor examines only 1–2 fields per case.

Two practical issues remain open. First, we tested on injected anomalies, not real fraud, and real anomalies are likely subtler. Second, even at the label-free operating point ($k^*_{\text{det}} = 9.5$) the 31 false positives still add some investigation overhead. BCE loss ranking can prioritize cases (Section V-H), and adjusting k_{det} (Table 7) lets auditors tune the recall–precision trade-off.

Real-world journal data would change CF performance in a few specific places.

Balanced entries. Real journal entries pair debit and credit lines that sum to zero. Our model handles one line at a time and so does not see balance at all. Adapting the pipeline would mean two things: (i) moving to an entry-level architecture, a set or sequence encoder over the lines sharing an accounting document number, so the AE can learn the balance regularity rather than treating each line in isolation, and (ii) adding the double-entry constraint $\sum_{\text{debit}} \text{DMBTR} = \sum_{\text{credit}} \text{DMBTR}$ as a *hard* constraint in CF search. The beam search already enforces per-attribute mutability constraints, so a balance constraint fits naturally as a feasibility filter that rejects any candidate mutation violating the sum, and the numeric gradient step (Section IV-G) can be projected onto the balance hyperplane after each update so that refined amounts stay balanced. This would keep every proposed correction a valid, balanced entry. A full entry-level treatment is beyond the present per-line scope but is a concrete next step.

TABLE 13. Complementarity of RESHAPE and CF.

	RESHAPE	CF
Question	Which attribute is the cause?	Which modifications would restore normality?
Exact match	—	25.0%
Partial hit	—	82.5%
Strength	Cause ranking	Actionable remediation
Weakness	Attribution to correlated attributes	May miss some causes due to minimal changes

More distinct categorical codes. With hundreds of HKONT codes and dozens of KTOSL/BSCHL codes, conditional distributions become sharper, and unseen-pair anomalies leave a more localized reconstruction-error signature. We expect exact match for unseen pairs (now 11% for KH and 17% for HP) to rise, because clearing one of the two attributes would more often leave a residual violation, and beam search would then have to touch both. The cost is a larger search space and slower runtime; the 4-attribute combinatorial wall in Section VI-F arrives sooner.

Categorical×amount dependence. The 0% exact match for unusual categ+amount comes from DMBTR being nearly uniform across HKONT codes (Section VI-C). When different HKONT codes have clearly different amount distributions, the AE learns that an HKONT code implies an amount range, and the CF can then propose a DMBTR correction when the amount is the real cause, and the gradient module that Section V-G validates in isolation would then engage on real anomalies. We expect this to move overall exact match more than the other two.

H. COMPUTATIONAL COST AND SCALABILITY

RESHAPE's SHAP calculation dominates computation time. KernelExplainer evaluates the AE 500 times per explained sample (100 background samples with 500 perturbations), so cost scales linearly with the number of flagged anomalies; by contrast, beam search with gradient refinement averages only 0.06 s per counterfactual (Table 9), since it needs only AE forward passes. For large-scale deployment the SHAP stage is therefore the bottleneck to attack. Amortized explainers replace the per-sample sampling with a single pass: FastSHAP [28] trains an explainer network once and then emits Shapley-value estimates in one forward pass per sample, cutting the per-sample cost from 500 model evaluations to one (about a 500× reduction in model calls at inference, traded for a one-time training cost); DeepSHAP/GradientSHAP [5] similarly replace sampling with a single backpropagation pass, and TreeSHAP [29] applies in closed form if a tree surrogate is fit to the reconstruction error. Because RESHAPE supplies only a search *priority*, an approximate attribution is sufficient, because the beam search corrects any mis-ranking, so these faster approximators are useful substitutes when the anomaly volume is high. Reducing the background sample size or perturbation count trades attribution fidelity for speed along the same axis.

VII. CONCLUSION

We presented a pipeline for explainable anomaly detection in accounting journal entries. It pairs RESHAPE (attribute-level SHAP attribution) for root-cause identification with RESHAPE-guided beam search and gradient-based numeric optimization for counterfactual generation. Detection uses a global BCE loss threshold selected label-free from a clean-validation false-positive-rate target; CF validity is checked per attribute.

On 532,789 normal SAP-format journal entries split into disjoint training, validation, and test partitions, the detector found all 120 held-out injected anomalies at the label-free operating point $k^*_{\mathrm{det}} = 9.5$, with 31 false positives on the clean test partition (FPR 3.9×10^{-4}). On those 120 anomalies, the CF pipeline exactly matched the ground truth in 30 cases (25.0%) and hit at least one ground-truth attribute in 99 cases (82.5%), and it outperformed the Wachter, DiCE, and FACE baselines on exact match, validity, and sparsity at once. Mean CF sparsity is about 1.18, so each CF typically changes one attribute and the correction shown to the auditor is short. Ablations confirm that RESHAPE prioritization drives the exact-match gain, and a controlled probe confirms the gradient module normalizes genuinely anomalous numeric attributes, even though this dataset's absent categorical×amount dependence keeps that path dormant for the injected anomalies.

ACKNOWLEDGMENT

The authors would like to thank Prof. Shinichi Shirayama, a Certified Public Accountant and former partner at Deloitte Touche Tohmatsu LLC, for fruitful discussions and for reviewing the content of this article from a professional auditing perspective.

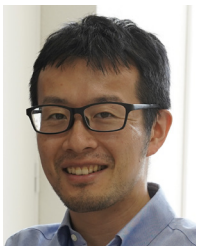
DECLARATION OF GENERATIVE AI

The authors used Claude Opus 4.8 (Anthropic) for coding assistance in the comparison with existing CF methods (Section V-E) and to improve the readability of the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

REFERENCES

[1] M. Schreyer, T. Sattarov, D. Borth, A. Dengel, and B. Reimer, "Detection of anomalies in large scale accounting data using deep autoencoder networks," 2017, *arXiv:1709.05254*.
[2] M. Schultz and M. Tropmann-Frick, "Autoencoder neural networks versus external auditors: Detecting unusual journal entries in financial statement audits," in *Proc. Annu. Hawaii Int. Conf. Syst. Sci.*, 2020, pp. 5421–5430.

[3] M. Schreyer, T. Sattarov, C. Schulze, B. Reimer, and D. Borth, "Detection of accounting anomalies in the latent space using adversarial autoencoder neural networks," in *Proc. KDD Workshop Anomaly Detection Finance*, 2019.
[4] A.-H. Karimi, G. Barthe, B. Schölkopf, and I. Valera, "A survey of algorithmic recourse: Contrastive explanations and consequential recommendations," *ACM Comput. Surveys*, vol. 55, no. 5, pp. 1–29, May 2023.
[5] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 4765–4774.
[6] R. Müller, M. Schreyer, T. Sattarov, and D. Borth, "RESHAPE: Explaining accounting anomalies in financial statement audits by enhancing SHapley additive exPlanations," in *Proc. 3rd ACM Int. Conf. AI Finance*, Nov. 2022, pp. 174–182.
[7] M. Schreyer, T. Sattarov, and D. Borth, "Multi-view contrastive self-supervised learning of accounting data representations for downstream audit tasks," in *Proc. 2nd ACM Int. Conf. AI Finance*, Nov. 2021, pp. 1–8.
[8] M. Schreyer, T. Sattarov, and D. Borth, "Federated and privacy-preserving learning of accounting data in financial statement audits," in *Proc. 3rd ACM Int. Conf. AI Finance*, Nov. 2022, pp. 105–113.
[9] H. Hemati, M. Schreyer, and D. Borth, "Continual learning for unsupervised anomaly detection in continuous auditing of financial accounting data," in *Proc. AAAI Workshop AI Financial Services*, 2022.
[10] T. Sattarov, D. Herurkar, and J. Hees, "Explaining anomalies using denoising autoencoders for financial tabular data," 2022, *arXiv:2209.10658*.
[11] H. Torabi, S. L. Mirtaheri, and S. Greco, "Practical autoencoder based anomaly detection by using vector reconstruction error," *Cybersecurity*, vol. 6, no. 1, p. 1, Jan. 2023.
[12] S. Verma, J. Dickerson, and K. Hines, "Counterfactual explanations and algorithmic recourses for machine learning: A review," 2020, *arXiv:2010.10596*.
[13] R. Guidotti, A. Monreale, F. Giannotti, D. Pedreschi, S. Ruggieri, and F. Turini, "Factual and counterfactual explanations for black box decision making," *IEEE Intell. Syst.*, vol. 34, no. 6, pp. 14–23, Nov. 2019.
[14] S. Wachter, B. Mittelstadt, and C. Russell, "Counterfactual explanations without opening the black box: Automated decisions and the GDPR," *Harvard J. Law Technol.*, vol. 31, no. 2, pp. 841–887, 2018.
[15] R. K. Mothilal, A. Sharma, and C. Tan, "Explaining machine learning classifiers through diverse counterfactual explanations," in *Proc. Conf. Fairness, Accountability, Transparency*, Jan. 2020, pp. 607–617.
[16] R. Poyiadzi, K. Sokol, R. Santos-Rodriguez, T. De Bie, and P. Flach, "FACE: Feasible and actionable counterfactual explanations," in *Proc. AAAI/ACM Conf. AI, Ethics, Soc.*, Feb. 2020, pp. 344–350.
[17] S. Sharma, J. Henderson, and J. Ghosh, "CERTIFAI: A common framework to provide explanations and analyse the fairness and robustness of black-box models," in *Proc. AAAI/ACM Conf. AI, Ethics, Soc.*, Feb. 2020, pp. 166–172.
[18] T. Laugel, M.-J. Lesot, C. Marsala, X. Renard, and M. Detyniecki, "Comparison-based inverse classification for interpretability in machine learning," in *Proc. Int. Conf. Inf. Process. Manage. Uncertainty Knowl.-Based Syst.*, 2018, pp. 100–111.
[19] K. Kanamori, T. Takagi, K. Kobayashi, and H. Arimura, "DACE: Distribution-aware counterfactual explanation by mixed-integer linear optimization," in *Proc. 29th Int. Joint Conf. Artif. Intell.*, Jul. 2020, pp. 2855–2862.
[20] A.-H. Karimi, B. Schölkopf, and I. Valera, "Algorithmic recourse: From counterfactual explanations to interventions," in *Proc. ACM Conf. Fairness, Accountability, Transparency*, Mar. 2021, pp. 353–362.
[21] S. Haldar, P. G. John, and D. Saha, "Reliable counterfactual explanations for autoencoder based anomalies," in *Proc. 3rd ACM India Joint Int. Conf. Data Sci. Manage. Data (8th ACM IKDD CODS 26th COMAD)*, Jan. 2021, pp. 83–91.
[22] D. Sulem, M. Donini, M. B. Zafar, F.-X. Aubet, J. Gasthaus, T. Januschowski, S. Das, K. Kenthapadi, and C. Archambeau, "Diverse counterfactual explanations for anomaly detection in time series," 2022, *arXiv:2203.11103*.
[23] L. Antwarg, R. M. Miller, B. Shapira, and L. Rokach, "Explaining anomalies detected by autoencoders using Shapley additive explanations," *Expert Syst. Appl.*, vol. 186, Dec. 2021, Art. no. 115736.
[24] Z. Li, Y. Zhu, and M. Van Leeuwen, "A survey on explainable anomaly detection," *ACM Trans. Knowl. Discovery Data*, vol. 18, no. 1, pp. 1–54, 2024.
[25] C. A. Zhang, S. Cho, and M. Vasarhelyi, "Explainable artificial intelligence (XAI) in auditing," *Int. J. Accounting Inf. Syst.*, vol. 46, Sep. 2022, Art. no. 100572.
[26] J. Černevičienė and A. Kabašinskas, "Explainable artificial intelligence (XAI) in finance: A systematic literature review," *Artif. Intell. Rev.*, vol. 57, no. 8, Jul. 2024, Art. no. 216.
[27] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, "LOF: Identifying density-based local outliers," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, 2000, pp. 93–104.
[28] N. Jethani, M. Sudarshan, I. C. Covert, S.-I. Lee, and R. Ranganath, "FastSHAP: Real-time Shapley value estimation," in *Proc. Int. Conf. Learn. Represent.*, 2022.
[29] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, "From local explanations to global understanding with explainable AI for trees," *Nature Mach. Intell.*, vol. 2, no. 1, pp. 56–67, Jan. 2020.



YOHEI KAWAZURA received the B.Eng., M.S., and Ph.D. degrees from The University of Tokyo, Japan, in 2008, 2010, and 2013, respectively. He was a Research Associate with the Graduate School of Frontier Sciences, The University of Tokyo, from 2014 to 2016, and a Postdoctoral Research Assistant with the Rudolf Peierls Centre for Theoretical Physics, University of Oxford, from 2016 to 2019. From 2019 to 2024, he was an Assistant Professor with the Frontier Research Institute for Interdisciplinary Sciences, Tohoku University. Since 2024, he has been an Associate Professor with the School of Data Science and Management, Utsunomiya University. His research interests include plasma physics and astrophysics, particularly magnetohydrodynamic and kinetic turbulence in accretion disks and the solar wind, as well as high-performance computational physics.



DAISUKE SAITOU received the B.Com. and M.Com. degrees from Meiji University, Japan, in 1995 and 1997, respectively. In 1997, he joined TKC Corporation, where he was engaged in the design and development of accounting systems and internal audit support systems. Since 2025, he has been an Associate Professor with the Center for Data Science, Utsunomiya University. His research interest includes accounting fraud detection.